

## Lab Exercises 3

### OUTLINE

- Example: Arranging functions in ascending order of growth rate
- Theorem 1: For every  $b > 1$  and every  $x > 0$ ,  $\log_b n = O(n^x)$
- Theorem 2: For every  $r > 1$  and every  $d > 0$ ,  $n^d = O(r^n)$
- Problem 1 (10 points): Classify functions using Big-O, Omega, and Theta notation
- Problem 2 (10 points): Arrange functions in ascending order of growth rate
- Problem 3 (10 points): True or False questions about Big-O notation
- Problem 4 (10 points): Prove that  $n^2 = O(2^n)$
- Problem 5 Assignments on OJ platform

#### IMPORTANT SUBMISSION INSTRUCTIONS:

- Problems 1-4: Submit on Canvas
- Problem 5: Complete on the OJ platform
- All submissions due by September 26th

**Example.** Take the following list of functions and arrange them in ascending order of growth rate. That is, if function  $g(n)$  immediately follows function  $f(n)$  in your list, then it should be the case that  $f(n)$  is  $O(g(n))$ .

1.  $f_1(n) = 10^n$
2.  $f_2(n) = n^{1/3}$
3.  $f_3(n) = n^n$
4.  $f_4(n) = \log_2 n$
5.  $f_5(n) = 2^{\sqrt{\log_2 n}}$

**Theorem 1:** For every  $b > 1$  and every  $x > 0$ , we have  $\log_b n = O(n^x)$

**Proof:** Recall that  $\log_b n = \frac{\ln n}{\ln b}$ , where  $\ln$  is the natural logarithm. Thus, we have:

$$\log_b n = \frac{\ln n}{\ln b}$$

We know that for any  $x > 0$ , the function  $n^x$  grows faster than  $\ln n$  as  $n$  becomes large. Specifically, the limit:

$$\lim_{n \rightarrow \infty} \frac{\ln n}{n^x} = 0$$

This means that for sufficiently large  $n$ ,  $\ln n$  is dominated by  $n^x$ .

Since  $\lim_{n \rightarrow \infty} \frac{\ln n}{n^x} = 0$ , there exists a constant  $C' > 0$  and an  $n_0$  such that for all  $n \geq n_0$ :

$$\frac{\ln n}{n^x} \leq C'$$

Multiplying both sides by  $n^x$  and then by  $\frac{1}{\ln b}$  (which is positive since  $b > 1$ ), we obtain:

$$\log_b n = \frac{\ln n}{\ln b} \leq \frac{C' \cdot n^x}{\ln b}$$

Let  $C = \frac{C'}{\ln b}$ . Then:

$$\log_b n \leq C \cdot n^x$$

We have shown that there exist constants  $C > 0$  and  $n_0$  such that for all  $n \geq n_0$ ,  $\log_b n \leq C \cdot n^x$ . Therefore,  $\log_b n = O(n^x)$ .

**Theorem 2:** For every  $r > 1$  and every  $d > 0$ , we have  $n^d = O(r^n)$ .

**Proof:** For  $r > 1$  and  $d > 0$ , the exponential function  $r^n$  grows faster than the polynomial function  $n^d$  as  $n$  becomes large. Specifically, the limit:

$$\lim_{n \rightarrow \infty} \frac{n^d}{r^n} = 0$$

This means that for sufficiently large  $n$ ,  $n^d$  is dominated by  $r^n$ .

Since  $\lim_{n \rightarrow \infty} \frac{n^d}{r^n} = 0$ , there exists a constant  $C > 0$  and an  $n_0$  such that for all  $n \geq n_0$ :

$$\frac{n^d}{r^n} \leq C$$

Multiplying both sides by  $r^n$ , we obtain:

$$n^d \leq C \cdot r^n$$

We have shown that there exist constants  $C > 0$  and  $n_0$  such that for all  $n \geq n_0$ ,  $n^d \leq C \cdot r^n$ . Therefore,  $n^d = O(r^n)$ .

**Solution:** We can deal with functions  $f_2$ , and  $f_4$  very easily, since they belong to the basic families of polynomials and logarithms. In particular, by Theorem 1, we have  $f_4(n) = O(f_2(n))$ . By Theorem 2, polynomial functions like  $n^{1/3}$  are  $O(\text{exponential functions})$  like  $10^n$ , so  $f_2(n) = O(f_1(n))$ .

For  $f_1(n) = 10^n$  and  $f_3(n) = n^n$ , we need to compare their growth rates. For large  $n$ ,  $n^n$  grows faster than  $10^n$ . This can be seen by taking the ratio:  $\frac{n^n}{10^n} = (\frac{n}{10})^n$ . As  $n$  grows larger than 10, this ratio increases without bound, showing that  $f_1(n) = O(f_3(n))$ .

Finally, we come to function  $f_5$ , which is admittedly kind of strange-looking. A useful rule of thumb in such situations is to try taking logarithms to see whether this makes things clearer. In this case,  $\log_2 f_5(n) = \sqrt{\log_2 n} = (\log_2 n)^{1/2}$ .  $\log_2 f_4(n) = \log_2(\log_2 n)$ , while  $\log_2 f_2(n) = \frac{1}{3} \log_2 n$ . All of these can be viewed as functions of  $\log_2 n$ , and so using the notation  $z = \log_2 n$ , we can write

$$\begin{aligned} \log_2 f_2(n) &= \frac{1}{3}z \\ \log_2 f_4(n) &= \log_2 z \\ \log_2 f_5(n) &= z^{1/2} \end{aligned}$$

Now it's easier to see what's going on. First, for  $z \geq 16$ , we have  $\log_2 z \leq z^{1/2}$ . But the condition  $z \geq 16$  is the same as  $n \geq 2^{16} = 65536$ ; thus once  $n \geq 2^{16}$  we have  $\log_2 f_4(n) \leq \log_2 f_5(n)$ , and so  $f_4(n) \leq f_5(n)$ . Thus we can write  $f_4(n) = O(f_5(n))$ .

Similarly we have  $z^{1/2} \leq \frac{1}{3}z$  once  $z \geq 9$  (once  $n \geq 2^9 = 512$ ). For  $n$  above this bound we have  $\log_2 f_5(n) \leq \log_2 f_2(n)$  and hence  $f_5(n) \leq f_2(n)$ , and so we can write  $f_5(n) = O(f_2(n))$ . Essentially, we have discovered that  $2^{\sqrt{\log_2 n}}$  is a function whose growth rate lies somewhere between that of logarithms and polynomials.

The final list:  $f_4 \leq f_5 \leq f_2 \leq f_1 \leq f_3$

**Problem 1 (10 points).** For each row  $i$  in Table 1, determine whether  $A_i$  belongs to  $O(B_i)$ ,  $\Omega(B_i)$ , or  $\Theta(B_i)$ . Place a checkmark (✓) in the appropriate column(s). The first two rows are provided as examples. No explanation is required.

Table 1: Big-O, Omega, and Theta Classification

| $A$               | $B$           | $O$ | $\Omega$ | $\Theta$ |
|-------------------|---------------|-----|----------|----------|
| $5n$              | $n$           | ✓   | ✓        | ✓        |
| $5$               | $n$           | ✓   |          |          |
| $n^3$             | $n^2$         |     |          |          |
| $\log n^2$        | $n$           |     |          |          |
| $3^n$             | $2^n$         |     |          |          |
| $\log(n!)$        | $n \log n$    |     |          |          |
| $\sqrt{n}$        | $n^{0.5}$     |     |          |          |
| $(\frac{1}{2})^n$ | $1$           |     |          |          |
| $n^{1.5}$         | $n(\log n)^2$ |     |          |          |
| $\log_2 n$        | $\ln n$       |     |          |          |
| $4^n$             | $2^{2n}$      |     |          |          |
| $n$               | $\log(n^n)$   |     |          |          |
| $100n + 50$       | $n$           |     |          |          |

**Problem 2 (10 points).** Consider the following list of functions. Arrange them in ascending order of growth rate. If function  $g(n)$  immediately follows function  $f(n)$  in your list, then it should be the case that  $f(n) = O(g(n))$ .

1.  $f_1(n) = n^{2.5}$
2.  $f_2(n) = \sqrt{2^n}$
3.  $f_3(n) = n + 10$
4.  $f_4(n) = 10n$
5.  $f_5(n) = 100n$
6.  $f_6(n) = n^2 \log n$

**Requirements:**

- Provide your final ordered list
- For each consecutive pair  $(f_i, f_j)$ , briefly justify why  $f_i = O(f_j)$
- Consider asymptotic behavior as  $n \rightarrow \infty$

**Problem 3 (10 points).** True or False? No explanation is required.